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Question 1 [C03] [10 Points]

Given the following classes, design the derived class **BarcelonaPlayer** to generate the expected output. Total Salary = (Goals × Goal Rate) + (Man of the Match Bonus if applicable). Consider 500 Tk per goal and 1000 Tk as bonus. If the salary is greater than 1000 tk, then a tax of 10% is applied.

Driver code	Output
<pre> public class PlayerTester { public static void main(String[] args) { BarcelonaPlayer p1 = new BarcelonaPlayer("Messi", 10, 36); p1.scoreGoal(); p1.scoreGoal(); p1.setManOfTheMatch(true); p1.playerInfo(); System.out.println("1-----"); p1.calculateSalary(); System.out.println("2-----"); p1.payTax(); System.out.println("====="); BarcelonaPlayer p2 = new BarcelonaPlayer("Lamine Yamal", 27, 17); p2.scoreGoal(); p2.setManOfTheMatch(false); p2.playerInfo(); System.out.println("3-----"); p2.calculateSalary(); System.out.println("4-----"); p2.payTax(); } } </pre>	<pre> Name: Messi Jersey Number: 10 Age: 36 Goals: 2 1----- Salary: 2000 TK 2----- Tax Paid: 200.0 TK ===== Name: Lamine Yamal Jersey Number: 27 Age: 17 Goals: 1 3----- Salary: 500 TK 4----- No Tax Required </pre>
Parent Class	
<pre> public class FootballPlayer { public String name; public int jerseyNumber, age, salary; public int goals = 0; public boolean isManOfTheMatch = false; public FootballPlayer(String name, int jerseyNumber, int age) { this.name = name; this.jerseyNumber = jerseyNumber; this.age = age; } public void playerInfo() { System.out.println("Name: " + name + "\nJersey Number: " + jerseyNumber + "\nAge: " + age + "\nGoals: " + goals); } } </pre>	

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Question 1 [C03] [10 Points]

Given the following class, design the derived class **BangladeshPlayer** to calculate total runs scored in a match. If total run is greater than 15, then the player wins a car. Additionally, if the age is greater than 30 and total runs scored is greater or equal to 15 then the player is promoted to senior player.

Driver code	Output
<pre> public class PlayerTester { public static void main(String[] args) { BangladeshPlayer p1 = new BangladeshPlayer("Tamim", 28, 40); p1.run(2); p1.fourboundary(); p1.sixboundary(); p1.playerInfo(); System.out.println("1-----"); p1.result(); System.out.println("2-----"); p1.promotion(); System.out.println("====="); BangladeshPlayer p2 = new BangladeshPlayer("Shakib", 75, 38); p2.sixboundary(); p2.sixboundary(); p2.fourboundary(); p2.playerInfo(); System.out.println("3-----"); p2.result(); System.out.println("4-----"); p2.promotion(); } } </pre>	<pre> Name: Tamim Jersey Number: 28 Age: 40 runs: 12 1----- Total Runs: 12 Result: Tamim lost 2----- Tamim lacks experience ===== Name: Shakib Jersey Number: 75 Age: 38 runs: 16 3----- Total Runs: 16 Result: Shakib wins a car 4----- Shakib is promoted to senior player </pre>
Parent Class	
<pre> public class CricketPlayer { public String name; public int jerseyNumber, age; public int runs= 0; public CricketPlayer(String name, int jerseyNumber, int age) { this.name = name; this.jerseyNumber = jerseyNumber; this.age = age; } public void playerInfo() { System.out.println("Name: " + name + "\nJersey Number: " + jerseyNumber + "\nAge: " + age + "\nruns: " +runs); } } </pre>	